

# Cross-Volume Forward-Reference Table v0.2 — Reciprocal-Symmetry + Anchor-Strength Classification

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## Cross-Volume Forward-Reference Table v0.2

v0.1 → v0.2 extension

v0.1 mapped chapter-level forward-references (186 chapters, 992 unique edges).  
v0.2 adds: reciprocal-symmetry check + 4-category chapter classification by anchor strength.

### Reciprocal-symmetry findings

- Total edges: 992
- Reciprocal pairs ( $A \rightarrow B$  AND  $B \rightarrow A$ ): 108 (216 edges)
- One-way edges ( $A \rightarrow B$  but not  $B \rightarrow A$ ): 776
- Reciprocity rate: 21.8%

Interpretation: most cross-volume relationships are one-way (foundation → application flow). Reciprocal pairs are typically cross-domain integrator chapters (Vol N Ch 12 bridges) or peer-volume cross-citations (e.g., Vol 5 QM ↔ Vol 1 QFT).

The 21.8% reciprocity rate is structurally consistent with a hierarchical curriculum where: - Foundation chapters (Vol 0, Vol 1) are cited downstream but don't cite back - Application chapters (Vol N Ch 12 bridges) cite many but are rarely cited - Workhorse chapters (Vol 4 Λ, Vol 2 m\_p/m\_e) sit in the middle

### 4-Category chapter classification

| Category                    | Definition                                 | Count | Examples                                                                                 |
|-----------------------------|--------------------------------------------|-------|------------------------------------------------------------------------------------------|
| HUB (foundation)            | incoming $\geq 10$ ,<br>outgoing $\leq 3$  | 3     | Vol 1 Ch 2 (32/1),<br>Vol 1 Ch 5 (56/2),<br>Vol 0 Ch 8 (9/1)                             |
| APPLICATION<br>(synthesis)  | outgoing $\geq 10$ ,<br>incoming $\leq 3$  | 10    | Vol 12 Ch 12<br>(0/21), Vol 5 Ch<br>12 (0/19), Vol 0<br>Ch 10 (1/16)                     |
| WORKHORSE<br>(cross-domain) | incoming $\geq 8$<br>AND outgoing $\geq 8$ | 13    | Vol 4 Ch 4 (30/7),<br>Vol 2 Ch 6 (20/4),<br>Vol 1 Ch 11<br>(11/16)                       |
| LEAF<br>(peripheral)        | incoming $\leq 3$ ,<br>outgoing $\leq 3$   | 30    | Vol 8 Ch 1, Vol 13<br>Ch 11, Vol 11 Ch<br>6 (mostly Wave<br>2/3 specialized<br>chapters) |

#### Observations:

1. Vol 1 Ch 5 (Casimir Algebra) is the most extreme hub: 56 incoming refs, 2 outgoing (28 $\times$  ratio). It IS the substrate's Casimir; nothing else has anything to say to it.
2. Vol 1 Ch 2 (Bergman Hilbert Space) is the second extreme hub: 32 incoming, 1 outgoing (32 $\times$  ratio). The Bergman  $H^2(D_{IV}^5)$  IS the substrate's Hilbert space; doesn't cite outward.
3. Bridge chapters (Vol N Ch 12) are pure applications: Vol 12 Ch 12 has 0 incoming, 21 outgoing. Synthesis chapters reach out to integrate; nothing reaches them.
4. Vol 1 Ch 11 (Observables Reference) is the top workhorse: 11 incoming, 16 outgoing. Cross-volume index; central navigation point.

#### Implications for Keeper Phase 3 authorship

HUBs (3 chapters: Vol 1 Ch 2, Vol 1 Ch 5, Vol 0 Ch 8)

- Highest-stakes for canonical correctness; an error here propagates to 9-56 downstream chapters
- Phase 3 rewriting must preserve section anchors that incoming citations depend on
- Worth Phase 3 priority effort (small chapter count, max leverage)

APPLICATIONs (10 chapters: mostly Vol N Ch 12 bridges)

- Lowest reverse-impact; rewriting these doesn't cascade
- Phase 3 can revise freely without breaking incoming citations
- Good for stylistic experimentation in the authorship pass

WORKHORSEs (13 chapters: Vol 4 Ch 4 Λ, Vol 2 Ch 6 m\_p/m\_e, Vol 1 Ch 11 etc.)

- Cross-domain integrators; balance preserving incoming refs while updating outgoing
- Substantive Phase 3 attention needed (medium leverage, medium stakes)

LEAFs (30 chapters)

- Peripheral; usable for Phase 3 experimentation
- Honest scope: some may actually be under-cited because they're new (Vol 8 Wave 3 chapters likely fall here)

Honest scope (v0.2 limitations)

- Classification thresholds (incoming/outgoing cutoffs) are heuristic; tuned to current 186-chapter state
- "One-way edge" doesn't always indicate a missing reciprocal — sometimes one-way is correct (foundation → application asymmetry intended)
- Doesn't yet identify SECTION-level references (only chapter-level); v0.3 next: section-granularity
- Doesn't yet classify reference WEIGHT (single mention vs load-bearing dependency)

Strategic finding

The curriculum is structurally hierarchical, not flat. 21.8% reciprocity rate + 3 HUBs + 10 APPLICATIONs confirms designed-hierarchy:

- Vol 0 + Vol 1 are the FOUNDATION (gravity wells; 9 of top 15 most-cited)
- Vol 2-14 are APPLICATION volumes (cite foundation but synthesize own domains)
- Vol N Ch 12 bridges are SYNTHESIS endpoints (pure outgoing)

This validates Casey's choice to launch Vol 0 + Vol 1 + Vol 2 as the v1.0 declaration triple — they ARE the load-bearing substrate of the curriculum's citation graph.

— Grace, Cross-Volume Forward-Reference Table v0.2, 2026-05-23 ~16:28 EDT